

Project Title

(Phishing Website Detection Using Random Forest and NLP-Based URL Analysis)

Machine Learning (25SC2107E)

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ABSTRACT

Phishing websites are fraudulent websites designed to steal sensitive user information such as usernames, passwords, banking credentials, and personal data by impersonating legitimate websites. As cyber threats continue to grow, identifying phishing websites quickly and accurately has become an important cybersecurity challenge.

This project proposes a Machine Learning-based phishing website detection system using the Random Forest algorithm combined with Natural Language Processing (NLP)-based URL analysis. The system analyzes various characteristics of URLs such as their length, presence of special characters, suspicious keywords, domain structure, token patterns, and lexical features extracted through NLP techniques. These extracted features are used to train a Random Forest classifier capable of distinguishing legitimate websites from phishing websites.

The proposed model aims to achieve high detection accuracy while maintaining low false-positive rates. The system can assist users by identifying potentially malicious websites before they access them, thereby reducing the risk of online fraud and identity theft.

The project demonstrates how Machine Learning and NLP techniques can be effectively integrated to improve cybersecurity applications and provide a reliable solution for phishing website detection.

Signature of the Guide